GEELON SO

Ph.D. Candidate, University of California, San Diego
Department of Computer Science and Engineering

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RESEARCH INTERESTS

Machine learning theory, sequential decision making, stochastic analysis, optimization, game theory

EDUCATION

University of California, San Diego — La Jolla, CA
Ph.D. Candidate, Computer Science — *Sep 2019—present*
Advisors: *Sanjoy Dasgupta & Yian Ma*

Columbia University — New York, NY
M.S. Computer Science — *May 2019*
Advisor: *Daniel Hsu*

The University of Chicago — Chicago, IL
B.S. Mathematics with Honors — *Jun 2017*
Advisor: *Stuart Kurtz*

PUBLICATIONS

Learnable mixed Nash equilibria are collectively rational.
with Yi-An Ma. Preprint, 2025.

Learning halfspaces without synthesized data.
with Hadley Black, Kasper Green Larsen, Arya Mazumdar, and Barna Saha. Preprint, 2025.

On the sample complexity of semi-supervised multi-objective learning.
with Tobias Wegel, Junhyung Park, and Fanny Yang. NeurIPS, 2025.

Preference optimization on Pareto sets: On a theory of multi-objective optimization.
with Abhishek Roy and Yi-An Ma. NeurIPS, 2025.

Consistency of k_n-nearest neighbor under adaptive sampling.
with Robi Bhattacharjee and Sanjoy Dasgupta. NeurIPS, 2025.

Online consistency of the nearest neighbor rule.
with Sanjoy Dasgupta. NeurIPS 2024.

Metric learning from limited pairwise preference comparisons.
with Zhi Wang and Ramya Korlakai Vinayak. UAI 2024.

Convergence of online k-means.
with Sanjoy Dasgupta and Gaurav Mahajan. AISTATS 2022.

EXPERIENCE

Simons Institute for the Theory of Computing Berkeley, CA
Visiting graduate student *Sep 2024—Dec 2024*

- Visitor for the Special Year on Large Language Models and Transformers program.

Amazon — Supply Chain Optimization Technologies (SCOT) team Bellevue, WA
Applied Scientist Intern *Jun 2024—Sep 2024*

- Framed and led the first ML augmentation of SCOT's LITMUS fulfillment simulator, replacing purely mechanistic rules with a data-driven generative component for non-deterministic events (e.g., SLAM times).
- Designed a scalable generative modeling pipeline in PyTorch to simulate correlated package trajectories in parallel, matching aggregate behavior, and not just per-package realism.
- Shipped an MLP (minimum lovable product) integrated with LITMUS on AWS; on JFK8 (NYC) validation, improved a key accuracy metric by >15% vs. the prior baseline (team typically targets basis-point gains).
- Operationalized an open-ended problem and delivered production-ready package and docs, enabling continued iteration by a non-ML team.

Seekr (news search engine company) Carlsbad, CA
Machine learning research and engineering intern *Jun 2022—Jun 2023*

- Spearheaded the overhaul of the trending news labeling process; developed a human-in-the-loop system to enable rapid experimentation with diverse rules- and ML-methods for keyword extraction and labeling.
- Engineered a data exploration tool to help developers gain intuition for the underlying news/text data; created modular system, streamlining new and iterative refinement of label extraction methods.
- Built a user-friendly GUI for human evaluation of label quality.
- The incorporation of the redesign resulted in significantly more expressive and meaningful topic labels.

TEACHING

The Institute for Emerging CORE Methods in Data Science La Jolla, CA
Lecturer for the Foundations in Data Science high school summer program *Jul 2023—Aug 2023*

- Designed and presented lectures/homework for a module on linear algebra; director: *Rajiv Gandhi*

University of California, San Diego La Jolla, CA
Teaching assistant

- Machine Learning, *Yian Ma*, *Fall 2022, Winter 2024*; Machine Learning, *Sanjoy Dasgupta*, *Winter 2025, Fall 2025*; Probability and Statistics, *Sanjoy Dasgupta*, *Fall 2020*

Columbia University New York, NY
Teaching assistant

- Awarded fellowship; presented 15 hours of lectures on unsupervised learning techniques; designed 9 homework problems; taught over 100 hours during office hours and individual meetings
- Unsupervised Learning, *Nakul Verma*, *Summer 2018*; Machine Learning, *Nakul Verma*, *Summer 2018*; Graph Theory, *Tim Sun*, *Spring 2018*; Geographic Information Systems, *Michael Parrott*, *Fall 2017*

SKILLS

Python, PyTorch, C, SQL, technical writing, LaTeX

AWARDS

2020 UC San Diego Changemaker Challenge 1st Place
In the UC San Diego COVID-19 Contact Tracing Challenge, sponsored by XYO

2019 Andrew P. Kosoresow Memorial Award for Excellence in Teaching and Service
Awarded for outstanding contributions to teaching in the Department of Computer Science at Columbia University and exemplary service to the Department and its mission

SERVICE

Reviewing
ACM CSUR *2024-2025*; AISTATS *2022–2026*; ALT *2025-2026*; ICML *2025*; JOTA *2025*; NeurIPS *2023, 2025*

EnCORE Outreach Volunteer La Jolla, CA
Volunteer instructor

UC San Diego Diversity Fellowship Committee La Jolla, CA
Reviewer *Jan 2020, Jan 2021*

CSE/HDSI Ph.D. Visit Day La Jolla, CA
Coordinator for AI/ML group *Mar 2020, Mar 2021*

Google/UC San Diego ExploreCSR Mentorship Program San Diego, CA
Volunteer mentor *Oct 2019—Jun 2020*

- Designed/taught a computational thinking course for underserved students in computer science

Friends of Washington Park Chicago, IL
Volunteer mentor *Jan 2014—Jul 2017*

- Tutored 5th–8th grader students in an after-school program in local neighborhood of Hyde Park

TALKS

Paper talks and posters

- Symposium on Mathematical Foundations of Trustworthy Learning (ETH Zurich) *Oct 2025*
- EnCORE Collaboration Workshop (UCSD) *Sep 2025*
- Chicago Junior Theorists Workshop (Northwestern and TTIC) *Dec 2024*
- Simons Institute for the Theory of Computing (Berkeley) *Oct 2024*
- Signals, Information, and Algorithms Lab (MIT) *Mar 2024*
- Yale Theory Student Seminar (Yale) *Mar 2024*
- EnCORE Student Social (UCSD) *Feb 2024*
- Redwood Center for Theoretical Neuroscience (Berkeley) *Dec 2023*
- FODSI CCSI Student Posters (MIT) *Jun 2023*
- EnCORE Student Social (UCSD) *Mar 2023*